

Riccardo Zuliani

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SUMMARY

Machine Learning Engineer with a solid academic background and hands-on experience in deep learning, time series anomaly detection, and computer vision. Skilled in Python, PyTorch, and key ML frameworks, with a strong focus on applying data-driven solutions to real-world challenges. Experienced in developing end-to-end pipelines for data analysis, model training, and deployment. Passionate about AI applications in industrial and automotive contexts, with a proactive and research-oriented mindset.

EXPERIENCE

- **NAIS Engineering Srl** Bologna Italy
Machine Learning Engineer & Researcher Oct 2024 – Present
 - **Data Analysis:** Developed multiple web application to better visualize project results and analysis using Dash Plotly framework.
 - **Data Engineering:** Built multiple aggregated datasets depending on the level of depth level on Formula 1 engines. A single record is sampled every 0.005 seconds (5 Hz).
 - **Neural Network Modelling:** Develop and evaluate different anomaly detection models on time series datasets.
- **Ca' Foscari University of Venice** Venice Italy
Probability and Statistics Tutor Nov 2023 – Jan 2024
 - **Tutorials:** Delivered 10 classes on key topics in probability and statistics which helped the students to succeed their exam.
 - **Assistance:** Assisted 30 students with solving statistical problems both manually and using R.
 - **One-on-one support:** Provided one-on-one support to clarify statistical concepts and methodologies.

PROJECTS

- **Active Learning with Dynamical System:** Developed a novel Active Learning methodology that uses Graph Transduction Algorithm to select the most difficult sample over a subset to be added in the known set.
- **Silhouette-Based Space Carving:** Reconstructed the 3D shape of an object by capturing it from multiple angles during rotation and progressively removing empty space based on object silhouettes.
- **Video Classification with Convolutional Neural Network:** Developed a benchmark over a set of different CNNs over the 1M Sports Youtube Video dataset. The project is inspired by the original white paper.
- **Dash AutoML Benchmark:** Implemented a web application to test and evaluate the current SOTA AutoML algorithm over a set of Kaggle and OpenML dataset, for both regression and classification problem types.

EDUCATION

- **Ca' Foscari University of Venice** Venice Italy
Master of Science in Computer Science - Data management and analysis Jan 2022 – Jul 2024
 - **Thesis:** Active Learning with Dynamical System.
 - **Relevant Courses:** Artificial Intelligence, 3D Geometric Computer Vision, Statistical Inference.
- **Ca' Foscari University of Venice** Venice Italy
Bachelor of Science in Computer Science - Data Science Sep 2019 – Oct 2021
 - **Thesis:** Dash AutoML Benchmark.
 - **Relevant Courses:** Algorithms and data structures, Databases, Data Analysis, Predictive Analytics.

SKILLS

- **Core Languages:** Python, C++, SQL, R.
- **Programming Framework:** PyTorch, OpenCV, Scikit-learn, Huggingface, Pandas, NumPy, Matplotlib, Seaborn, Dash Plotly, Flask.
- **Technologies:** Deep Learning, Machine Learning, Computer Vision, Git, Linux, Docker.
- **Languages:** Italian (Native), English (B2+, Upper-Intermediate).